

Project Report

RETAIL SALES DATA CLEANING & VISUALIZATION

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Objectives:

Raw sales data from all stores is cleaned, analysed to find hidden patterns and trends, and visualized through simple charts. These insights improve stock management and boost sales. The process converts messy, unstructured data into informed, actionable business decisions that increase revenue.

Key Insights:

- The variance of revenue of over months is noticeable.
- After every month with high revenue revenue level falls down due to unavailability of stock.
- Some product categories give maximum part of revenue.
- Customers use one payment method, showing a clear preference for that option.
- Revenue increases to a certain percentage of discount then it decreases.

KPI Summary:

Matric	Value
Total Revenue	₹ 63,106
Total Orders	367
Average Order Value	₹ 171.95
Top City	Houston
Top Category	Clothing

Dashboard Summary:

A multi-panel dashboard was designed to present key performance indicators, including sales trends and product performance metrics, facilitating efficient data-driven decision-making.

Recommendations:

1. Increase marketing and stock level then revenue is on peak.
2. Focus the categories who are generate low revenue to boost revenue.
3. Increase advertisement to boost revenue in high performing cities.
4. Focus on logistics and supply chain of the cities with low revenue.
5. Increase latest and trendy fashion stock to attract new generation boost revenue from them.
6. Should apply moderate discount without increase profit margin.

Conclusion:

The analysis highlights clear customer behaviour patterns and sales trends that can help improve business performance. Data-driven strategies based on these insights can enhance revenue growth, customer satisfaction, and operational efficiency.

Business Impact:

The insights obtained from this analysis can help businesses optimize inventory planning, improve marketing strategies, and enhance customer targeting. Data-driven decisions can significantly increase profitability and operational efficiency.

Key Findings:

- Sales show seasonal patterns with peak months generating highest revenue.
- Clothing category dominates total sales.
- Adults form the largest customer segment.
- Credit Card payments are preferred by most customers.
- A few cities contribute majority of revenue.